

HALI SARAH PARSONS

Data Science | Bellevue University | Expected January 2027

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SUMMARY

Data science student (graduating January 2027) who turns messy data into insights businesses can act on. A year of project-based coursework in Python data wrangling, visualization, NLP, statistics, predictive modeling and business intelligence design. Comfortable owning a project from requirements to a finished deliverable.

DATA SCIENCE PROJECTS

Predicting Economic Damage from U.S. Storm Events Group, Oct 2026

- Analyzed 410,311 NOAA storm events (2020–2025) to predict whether an event causes economic damage; Random Forest reached ROC-AUC 0.880 and average precision 0.669 vs. a 0.219 no-skill baseline.
- My role: data visualization.** Built the charts comparing storm frequency with damage rates, showing common events (Heat, Hail) rarely carry recorded damage while Tornado and Strong Wind often do.

Custom Named Entity Recognition May 2026

- Built a spaCy EntityRuler pipeline with four custom entity labels, extracting 2,501 entities from 2,540 sentences across five full-text PubMed Central articles (64,006 words) pulled live through an API.
- Applied sentence-level VADER sentiment to analyze how health outcomes and age groups are framed.

Digital Habits vs. Mental Health Analysis May 2026

- Cleaned a 100,000-row dataset (removed 1,865 duplicates, flagged outliers with IQR, engineered sleep-quality categories) and built five visualizations on 98,135 records.
- Showed higher screen time tracks with lower mood across all three stress tiers.

BI Justification, Requirements & Architecture BI coursework

- Wrote a business case (objectives, scope, timeline, budget, TCO, ROI) and a detailed requirements document with stakeholder interview inputs for a digital wellness analytics platform.
- Designed an end-to-end data architecture covering ingestion, ETL, data lake and warehouse, ML and BI layers.

Invoice Data Extraction in Power Automate Apr 2026

- Trained and tested two custom AI Builder document-processing models (16 fields; 31 fields and 4 tables) and ran an automated flow successfully.

Additional Coursework Projects 2026

- Sentiment analysis and logistic regression with TF-IDF, feature engineering and LDA topic modeling, Universal Sentence Encoder embeddings, and Bayes / Shannon entropy problem sets in Python.

EDUCATION

Bellevue University

Data Science

Expected January 2027

TECHNICAL SKILLS

LANGUAGES

Python, SQL, R / R Markdown, HTML, CSS, JavaScript

LIBRARIES

pandas, NumPy, SciPy, scikit-learn, spaCy, NLTK, matplotlib, seaborn

ANALYTICS & ML

Statistics (Bayes, Central Limit Theorem, entropy), regression, logistic regression, random forest, feature engineering, TF-IDF, topic modeling, named entity recognition, sentiment analysis, deep learning embeddings

BI & AUTOMATION

BI requirements and justification, data architecture and ETL design, Power Automate (AI Builder), Zapier, Jupyter / Colab, graphic design

COURSEWORK

- Essential Mathematics for Data Science
- Practical Python Data Wrangling
- Natural Language Processing & Text Analytics
- Data Visualization with R
- Business Intelligence Design
- Predictive Modeling Capstone